

# Final project

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## Introduction

This report follows the required steps for a longitudinal social network analysis using stochastic actor-oriented models (SAOMs) in RSiena. We analyze directed “best friends” nominations collected in Dutch secondary school classrooms across multiple waves (waves  $\mathbf{V}$ ,  $\mathbf{X}$ ,  $\mathbf{Y}$ ). The classroom network and pupil attributes come from the dataset collected by Andrea Knecht as part of her dissertation research.

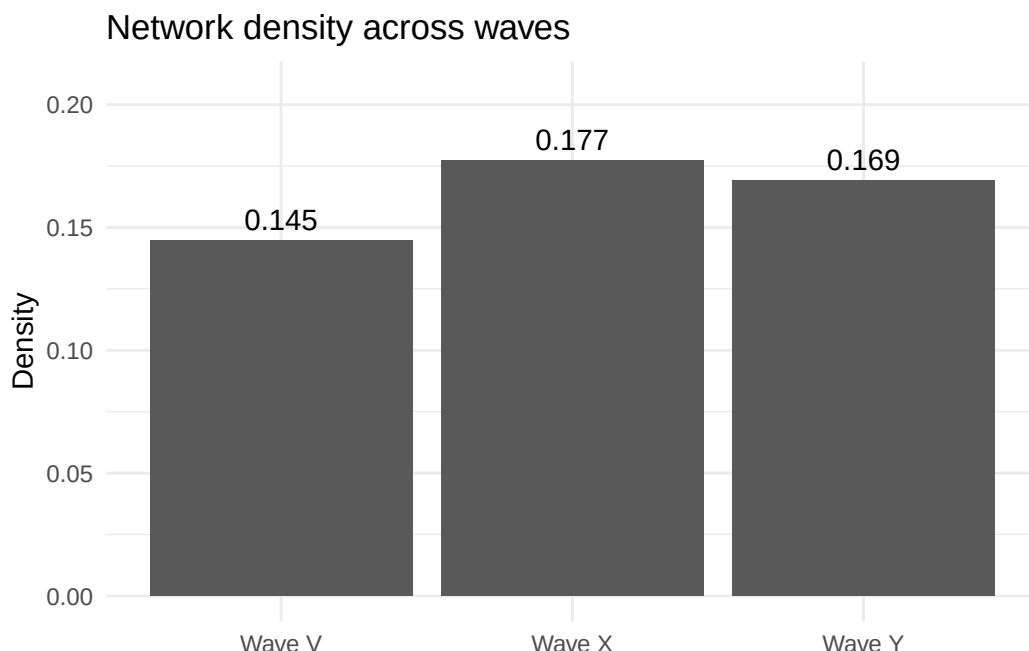
We examine whether (i) sex homophily shapes best-friend nominations (dyadic hypothesis), (ii) triadic closure/transitivity drives clustering (triadic hypothesis), and (iii) a node-level mechanism of popularity reinforcement (indegree-related popularity; inPop). We also include standard endogenous network controls (e.g., reciprocity and degree-related effects) and evaluate model fit using goodness-of-fit (GoF) simulations.

## 1. Network Descriptives

We focus on classroom **09f** (variable `schoolnr`). This classroom was selected because it provides at least three observed waves and shows sufficient stability (composition and tie Jaccard) to support SAOM estimation while still exhibiting meaningful change over time. Although the original dataset contains four survey waves (W, V, X, Y), we restrict the analysis to waves V, X, and Y. Including wave W would reduce the set of pupils observed in all waves (a constant node set is required for SAOM estimation), which would substantially shrink the network used in the longitudinal model. Therefore, we focus on the three waves that preserve sufficient overlap and comparability for SAOM estimation.

We generate a figure showing network density across waves.

Figure 1: Network density across waves



These values are proportions of possible friendships that exist. So for Wave V, only about 14.5% of all possible ties were actually reported. Wave X is slightly denser (about 17.7%), and Wave Y is very similar (about 16.9%). Networks with density  $< 0.2$  are considered sparse, which is very common in social networks because not everyone is friends with everyone else. We generate a table showing basic descriptives of the best-friends network across waves.

Table 1: Basic descriptives of the best-friends network across waves

| Wave   | Nodes | Density | DensityChangePct |
|--------|-------|---------|------------------|
| Wave V | 30    | 0.145   | NA               |
| Wave X | 28    | 0.177   | 22.386           |
| Wave Y | 26    | 0.169   | -4.524           |

Over the three waves of data collection, the selected classroom shows a moderate change in its social structure. The number of students decreased slightly from 30 in Wave 1 to 28 in Wave 2, and then to 26 in Wave 3, indicating some turnover in classroom composition. Density increases from 0.145 (Wave V) to 0.177 (Wave X), and then slightly decreases to 0.169 (Wave Y). Composition is relatively stable (Jaccard node-set: 0.933 for  $V \rightarrow X$  and 0.862 for  $X \rightarrow Y$ ), which supports SAOM modeling on a constant node set. However, tie stability is lower between  $V \rightarrow X$  (tie Jaccard = 0.242) and moderate between  $X \rightarrow Y$  (0.470), indicating substantial turnover in best-friend nominations, which makes endogenous structural effects (e.g., reciprocity and transitivity) especially relevant. For the selected classroom, the “Best friends” network was constructed from the survey responses, transforming each student’s nominations into an adjacency matrix. Missing codes 98/99 (as documented in the codebook) were treated as missing nominations to allow the network to be fully analyzed. We generate tables showing (i) the Jaccard stability of the node set across waves and (ii) the number of students present in all three waves.

Table 2: Jaccard stability of the node set (classroom composition)

| Wave_Comparison             | Jaccard_NodeSet |
|-----------------------------|-----------------|
| Wave V $\rightarrow$ Wave X | 0.933           |
| Wave X $\rightarrow$ Wave Y | 0.862           |

Table 3: Number of students present in all three waves (intersection)

| Students_in_all_three_waves |
|-----------------------------|
| 25                          |

Table 4: Jaccard stability of ties on the constant node set (intersection sample)

| Wave_Comparison             | Jaccard_Ties |
|-----------------------------|--------------|
| Wave V $\rightarrow$ Wave X | 0.242        |
| Wave X $\rightarrow$ Wave Y | 0.470        |

The table above reports **tie Jaccard** stability (overlapping friendships) between consecutive waves on the constant node set used for SAOM. Higher values indicate more stable friendship relations across time, supporting the use of longitudinal modelling. The Jaccard indices above quantify how stable the classroom composition is across waves. Because SAOMs require a constant node set, the subsequent SAOM analysis will be run on the students who are present in all three waves (the intersection).

## 2. Dyadic Hypothesis

For the analysis of dyadic characteristics in the “Best friends” network, we formulated the following hypothesis: students are more likely to nominate as best friends those peers who are of the same sex. This hypothesis is based on the well-established principle of sex homophily in adolescent social networks, where boys tend to form friendships with other boys, and girls with other girls. Theoretical considerations, such as social identity theory, suggest that individuals prefer closer ties with peers who share salient characteristics like sex, which can also facilitate participation in similar classroom activities or interests. Empirical studies in classroom social networks consistently report high levels of sex-based homophily (e.g., Snijders et al., 2010). To test sex homophily in RSiena, we use the sameX effect with the actor covariate `sex_cov`. A positive sameX parameter indicates that pupils are more likely to nominate peers of the same sex as best friends. For descriptive illustration, we display a  $5 \times 5$  subset of the same-sex indicator matrix derived from `sex_cov` (1 = same sex; 0 = different sex). This matrix is shown only as a descriptive check; the SAOM uses `sameX(sex_cov)` directly and does not require a separate dyadic covariate.

Table 5: Sex similarity matrix (Wave V;  $5 \times 5$  subset)

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 1 |

Table 6: Sex similarity matrix (Wave X; 5×5 subset)

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |

Table 7: Sex similarity matrix (Wave Y; 5×5 subset)

|   | 1 | 2 | 3 | 4 | 6 |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 1 | 1 |
| 2 | 1 | 1 | 0 | 1 | 1 |
| 3 | 0 | 0 | 1 | 0 | 0 |
| 4 | 1 | 1 | 0 | 1 | 1 |
| 6 | 1 | 1 | 0 | 1 | 1 |

To operationalize this dyadic hypothesis, sex similarity matrices were constructed for each wave, where each cell represents whether a pair of students share the same sex (coded as 1) or not (coded as 0). Examples of these matrices for a subset of students are:

Wave V (subset 5×5) 1 2 3 4 5 1 1 1 0 1 0 2 1 1 0 1 0 3 0 0 1 0 1 4 1 1 0 1 0 5 0 0 1 0 1

Wave X (subset 5×5): 1 2 3 4 5 1 1 0 1 0 1 2 0 1 0 1 0 3 1 0 1 0 1 4 0 1 0 1 0 5 1 0 1 0 1

Wave Y (subset 5×5, new student 6 included): 1 2 3 4 6 1 1 1 0 1 1 2 1 1 0 1 1 3 0 0 1 0 0 4 1 1 0 1 1 6 1 1 0 1 1

These matrices can be used in stochastic actor-oriented models (SAOMs) to formally test the influence of sex similarity on the formation and maintenance of friendship ties across waves. The consistent pattern of higher values along clusters of same-sex students supports the theoretical expectation that sex homophily plays a role in the friendship formation in this classroom

### 3. Triadic Hypothesis

For the analysis of triadic characteristics in the “Best friends” network, I formulated the following hypothesis: friendship ties are likely to exhibit transitivity, meaning

that if student A nominates student B as a best friend, and student B nominates student C, then student A is also likely to nominate student C. This hypothesis is grounded in the principle of triadic closure, a common phenomenon in social networks, where “friends of friends tend to become friends” (Granovetter, 1973; Snijders et al., 2010). Theoretical explanations suggest that transitive ties increase trust, reduce social tension, and facilitate cooperation within a group. In classroom networks, transitivity is often observed because students share activities, classes, and social environments, making it more likely that friendship clusters form. To operationalize this triadic hypothesis in RSiena, we use the transTrip effect, which captures triadic closure directly within the SAOM. A positive transTrip parameter indicates that the presence of  $A \rightarrow B$  and  $B \rightarrow C$  increases the likelihood of  $A \rightarrow C$  forming over time, reflecting clustering in the classroom network. As a descriptive check, we also compute and report global transitivity (triadic closure) for each wave.

We generate a table showing global transitivity (triadic closure) for each wave.

Table 8: Global transitivity by wave

| Wave   | Transitivity |
|--------|--------------|
| Wave V | 0.479        |
| Wave X | 0.610        |
| Wave Y | 0.500        |

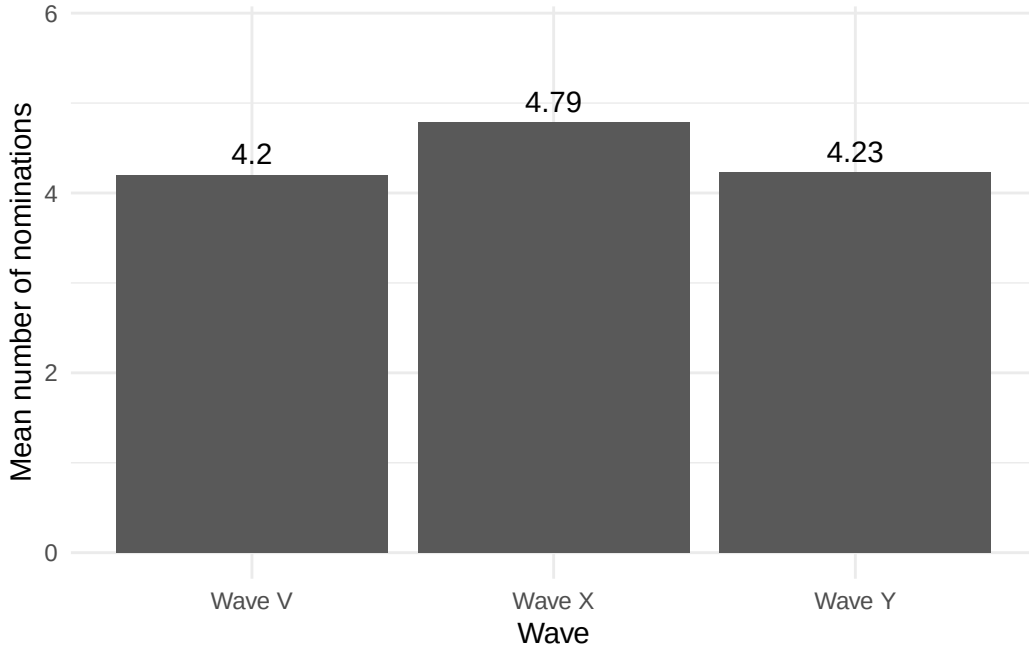
The transitivity of the “Best friends” network, which measures the tendency of friends-of-friends to also be friends (triadic closure), shows a moderate to high level of clustering across the three waves. In Wave V, the transitivity is 0.479, indicating that a substantial share of possible triads are closed. Wave X exhibits a transitivity of 0.61, reflecting the tendency for friendship triads to form. In Wave Y, the transitivity is 0.5, suggesting that the level of triadic closure remains in a similar range over time. These results support the hypothesis that friendship formation in the classroom is characterized by transitive relationships, consistent with established findings in adolescent social networks where social groups tend to form tightly connected clusters

#### 4. Node Attribute Hypothesis

For the analysis of node-level characteristics in the “Best friends” network, we formulated the following hypothesis: students with higher levels of popularity (in-degree) are more likely to be nominated as best friends by their peers. This hypothesis is based on the idea that individuals with high social status or visibility

in the classroom attract more friendship nominations, consistent with the principle of preferential attachment in social networks. Empirical studies in adolescent classrooms indicate that popular students tend to receive more friendship ties and maintain them over time (e.g., Snijders et al., 2010). To operationalize this hypothesis, a popularity score was assigned to each student based on the total number of received best-friend nominations (indegree). This score can then be used to capture popularity dynamics in a SAOM (here via the inPop effect) to test whether higher popularity increases the likelihood of receiving additional nominations over time.

Figure 2: Mean popularity (indegree) across waves.



For the node-level analysis, the popularity of each student was calculated as the total number of best-friend nominations they received in each wave. The data were initially transformed into a long format, where each student in each wave was represented as a separate row, resulting in 84 rows in total (30 students in Wave V, 28 in Wave X, and 26 in Wave Y). To summarize the key characteristics of this node-level variable, we computed the mean, minimum, and maximum popularity per wave. In Wave V, students received on average 4.20 nominations, with a range from 0 to 8. In Wave X, the mean popularity increased slightly to 4.79, with the same maximum of 8 and a minimum of 1. Wave Y showed a mean popularity of 4.35, with a slightly lower maximum of 7 and the same minimum of 1. These summaries indicate that while most students receive several friendship nominations,

some students are consistently more popular than others, which may influence the formation and maintenance of best-friend ties over time. (For SAOM estimation, we use a constant node set across waves as required by the model.)

## 5. Control Variables

Our analysis of the “Best friends” network, several control variables were included to account for factors that might influence friendship formation and maintenance:

Ego-level variables: These describe characteristics of the student sending the friendship nomination. For example, we include outdegree activity (`outAct`) to account for heterogeneity in pupils’ tendency to send nominations (i.e., some students nominate more peers than others).

Alter-level variables: These describe characteristics of the student receiving the nomination. For instance, we include indegree-related popularity (`inPop`) to capture whether pupils who already receive many nominations become more/less likely to receive additional nominations over time (an endogenous degree-related mechanism).

Dyadic variables: These variables describe pairwise relationships between students. For example, sex homophily is modeled via `sameX` using the actor covariate `sex_cov`, capturing the tendency for students to nominate same-sex peers as best friends.

Structural/network variables: These account for endogenous network processes, such as outdegree (`density`), which captures the baseline sparsity of the network, reciprocity, which captures the tendency for a nomination to be returned by the alter, and transitivity (`transTrip`), which measures the tendency for friends of friends to become friends. (If the updated specification is used as the final model, we also include `reciAct` as an additional endogenous control to improve model fit for degree distributions.)

Together, these control variables allow the stochastic actor-oriented model to isolate the effects of individual, dyadic, and structural factors on the evolution of the friendship network over time.

We generate a table listing the control variables used in the analysis..

Table 9: Control variables to be included in the analysis

| Variable                            | Type                              | Description                                                                                                                 |
|-------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Outdegree (density)                 | Structural (baseline)             | Baseline tendency toward a sparse nomination network.                                                                       |
| Reciprocity                         | Dyadic/structural (endogenous)    | Tendency to reciprocate a best-friend nomination.                                                                           |
| Sex homophily (sameX)               | Dyadic (based on actor covariate) | Preference for same-sex nominations (computed from sex_cov via sameX).                                                      |
| Transitivity (transTrip)            | Triadic/structural (endogenous)   | Friends-of-friends closure: $A \rightarrow B$ and $B \rightarrow C$ increases $A \rightarrow C$ .                           |
| Indegree-related popularity (inPop) | Degree-related / endogenous       | Whether higher-indegree pupils become more/less likely to receive additional nominations over time (endogenous popularity). |
| Outdegree activity (outAct)         | Degree-related / endogenous       | Heterogeneity in pupils' tendency to send nominations beyond baseline effects.                                              |
| Reciprocity activity (reciAct)      | Degree-related / endogenous       | Heterogeneity related to reciprocating nominations; added to improve outdegree fit (Model 2).                               |

## 6. Longitudinal Plots

We visualize the best-friends network over time (waves V, X, Y) using fixed node positions and node colors by popularity (indegree; number of nominations received). Popularity is recalculated in each wave on the constant node set used for SAOM. Fixed positions ensure that changes in ties and in node popularity across waves can be compared directly without layout-induced movement of nodes. Figures Figure 3, Figure 4, and Figure 5 visualize the best-friends network in Waves V, X, and Y using fixed node positions and node colors by popularity (indegree).

Figure 3: Best-friends network in Wave V (fixed node positions; node color indicates popularity/indegree).

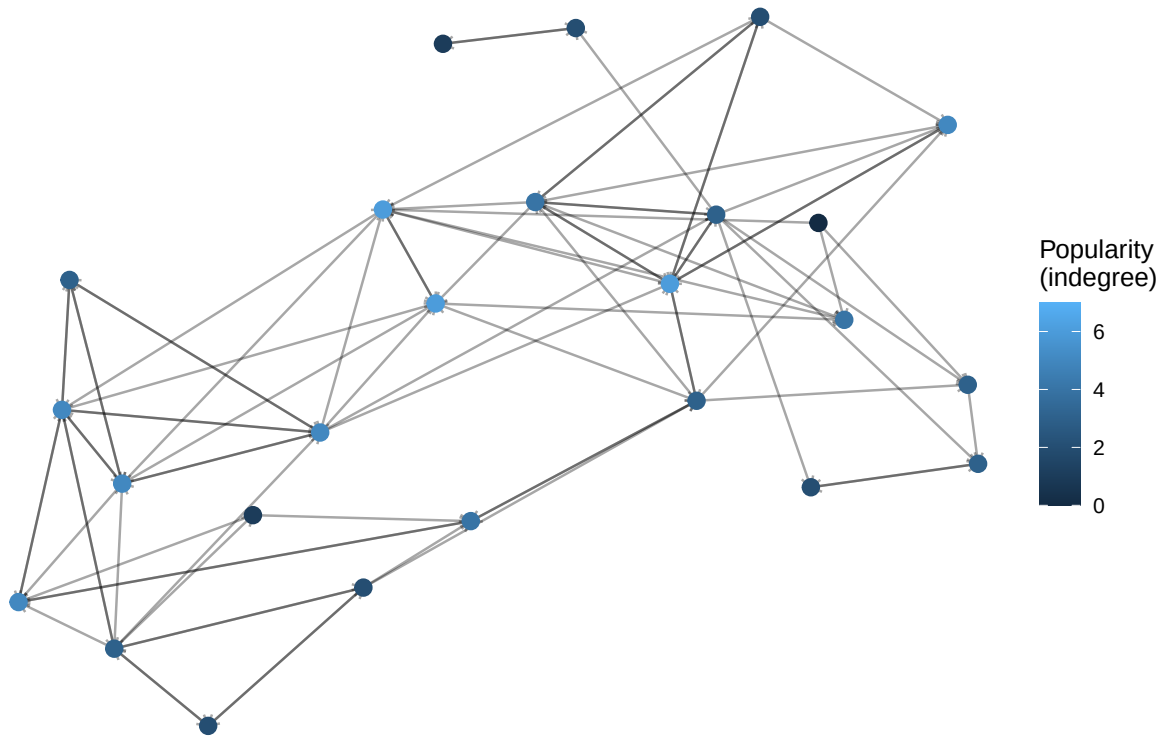


Figure 4: Best-friends network in Wave X (fixed node positions; node color indicates popularity/indegree).

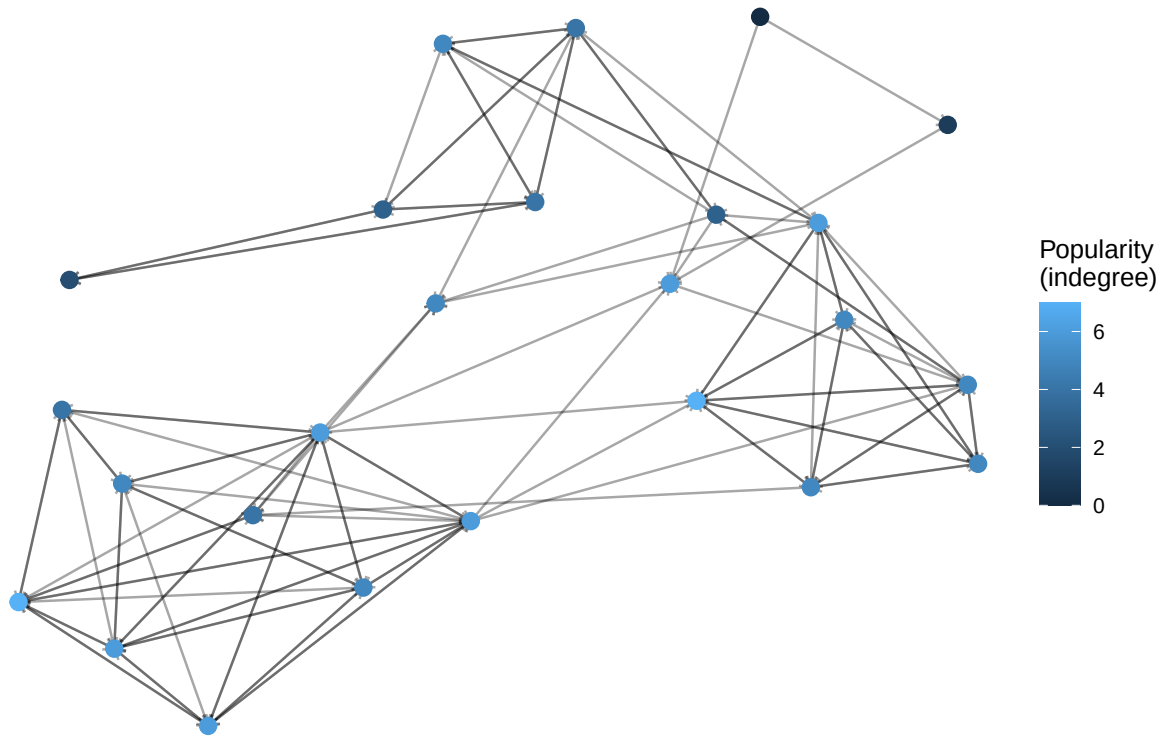
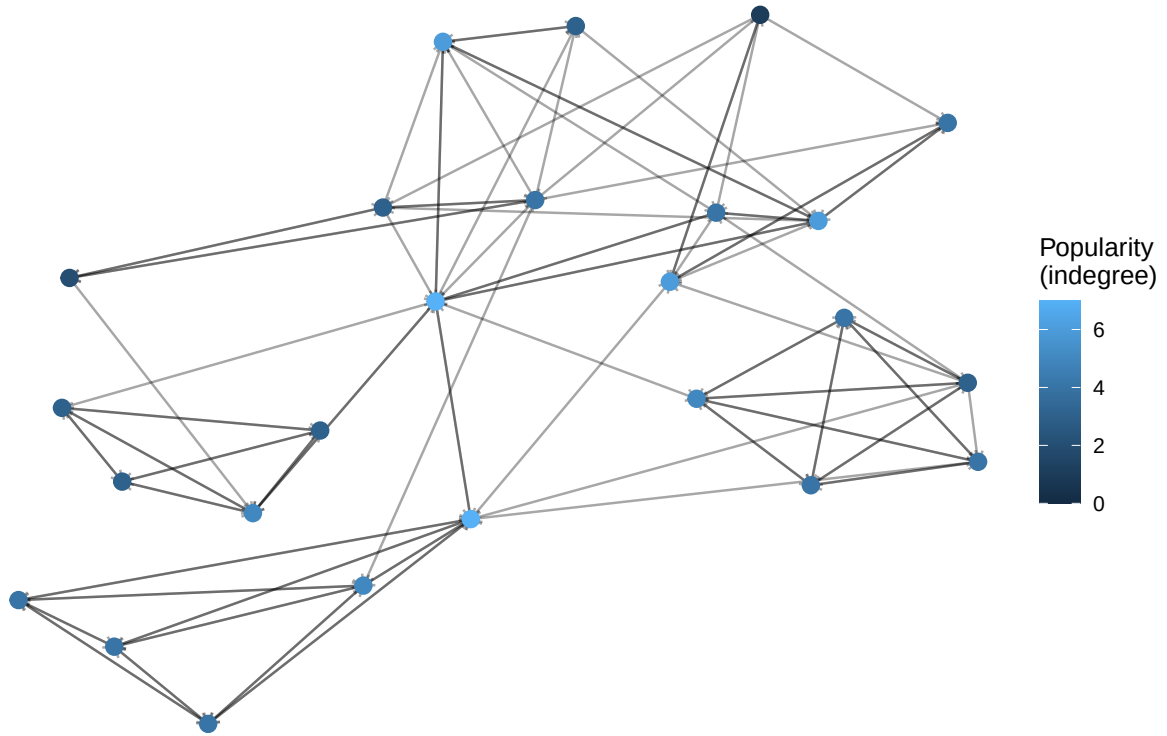


Figure 5: Best-friends network in Wave Y (fixed node positions; node color indicates popularity/indegree).



## 7. Model Specification

In this project, we test three hypotheses using a stochastic actor-oriented model (SAOM) in RSiena. We operationalize each hypothesis by selecting the corresponding RSiena effect(s). We generate a table showing how each hypothesis is translated into a SAOM effect.

Table 10: Operationalization of hypotheses as RSiena effects.

| Hypothesis                 | RSiena_effect                     | Expc_di-<br>rec | Interpretation                                                                                    |
|----------------------------|-----------------------------------|-----------------|---------------------------------------------------------------------------------------------------|
| H1 (Dyadic): sex homophily | sameX<br>(interaction1 = sex_cov) | Positive        | Ties are more likely between pupils of the same sex.                                              |
| H2 (Triadic): transitivity | transTrip                         | Positive        | Friend-of-a-friend closure: $A \rightarrow B$ and $B \rightarrow C$ increases $A \rightarrow C$ . |

| Hypothesis                                      | RSiena_effect                     | Expc_di-<br>rec | Interpretation                                                                                                 |
|-------------------------------------------------|-----------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------|
| H3 (Node-level):<br>popularity<br>reinforcement | inPop<br>(indegree<br>popularity) | Posi-<br>tive   | Pupils who already receive many nominations<br>are more likely to receive additional<br>nominations over time. |

First, sex homophily is modeled with the **sameX** effect using the sex covariate (`interaction1 = "sex_cov"`). A positive parameter indicates that pupils are more likely to nominate same-sex peers as best friends. Second, transitivity is captured by **transTrip**, where a positive parameter indicates triadic closure (“friends of friends become friends”). Third, our node-level hypothesis about popularity is operationalized with **inPop** (indegree popularity), an endogenous effect that tests whether pupils who already receive many nominations tend to attract additional nominations over time. Finally, we include **outAct** (outdegree activity) as an endogenous structural control to account for heterogeneity in pupils’ tendency to send nominations beyond the focal hypotheses.

## 8. Model Estimation

We conduct RSiena analyses to test the hypotheses above. Because SAOMs require the same node set across waves, we use the SAOM-aligned matrices (`netV_saom`, `netX_saom`, `netY_saom`) constructed from pupils present in all three waves. We estimate an initial model (Model 1) including our hypothesis-driven effects (**sameX**, **transTrip**, and **inPop**) plus a standard endogenous control for degree-related heterogeneity (**outAct**). If convergence is not reached, we increase estimation effort (`n3`) and continue from the previous estimates using `prevAns`.

We generate a table showing the estimated parameters of the final converged model.

Table 11: Convergence diagnostics for the final estimation run.

| Metric       | Value |
|--------------|-------|
| tconv.max    | 0.193 |
| max( tconv ) | 0.041 |

The Table 12 shows the parameter estimates for the final model.

Table 12: RSiena parameter estimates for the final model.

| Effect                                  | Estimate | SE    | t-ratio |
|-----------------------------------------|----------|-------|---------|
| outdegree (density)                     | -0.814   | 0.430 | -1.895  |
| reciprocity                             | 3.604    | 0.586 | 6.150   |
| transitive triplets                     | 0.466    | 0.052 | 8.988   |
| indegree - popularity                   | -0.336   | 0.079 | -4.274  |
| outdegree - activity                    | 0.127    | 0.043 | 2.929   |
| rec.degree <sup>^(1/1)</sup> - activity | -0.460   | 0.109 | -4.218  |
| same sex_cov                            | 0.246    | 0.124 | 1.981   |

## 9. Model Fit

Goodness-of-fit (GoF) assesses whether the fitted SAOM can reproduce important structural features of the observed network. We simulate multiple networks from the fitted model and compare observed network statistics to the distribution of the same statistics in the simulated networks. If the observed statistics are not extreme relative to the simulated distribution (and the Monte Carlo p-values are not close to 0 or 1), the model is considered to fit those features reasonably well. Here, we evaluate GoF for the indegree and outdegree distributions. We then update the model once (Model 2) by adding `reciAct` and re-run GoF to assess whether fit improves. Figure 6 and Figure 7 show the goodness-of-fit for indegree and outdegree in Model 1, and Table 13 summarizes the Monte Carlo p-values for Model 1 vs Model 2.

Figure 6: Goodness-of-fit for the indegree distribution (Model 1).

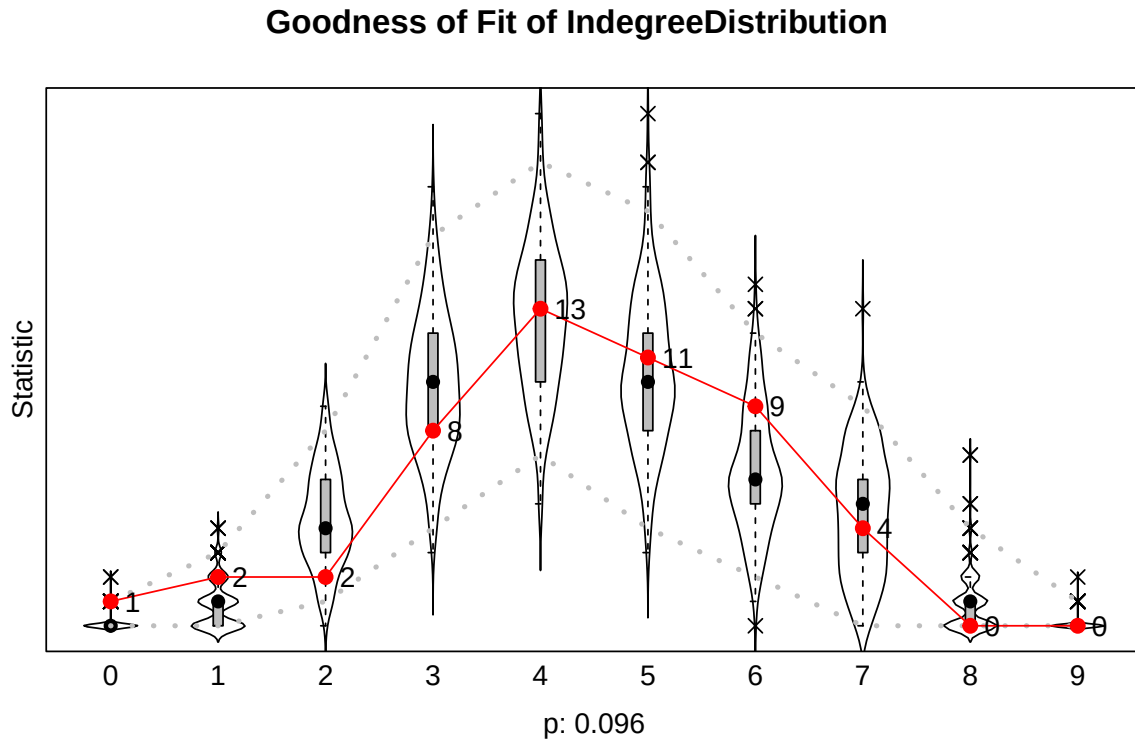


Figure 7: Goodness-of-fit for the outdegree distribution (Model 1).

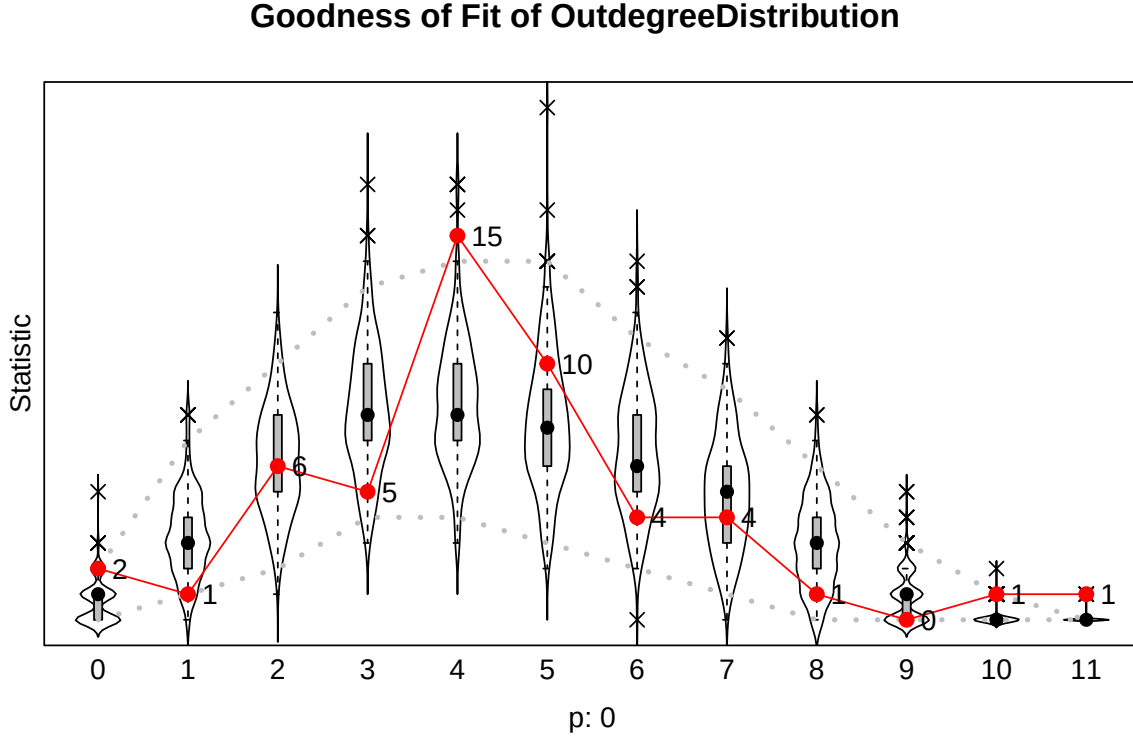


Table 13: Comparison of GoF Monte Carlo p-values (Model 1 vs Model 2).

| Statistic              | P_Model1 | P_Model2 |
|------------------------|----------|----------|
| Indegree distribution  | 0.096    | 0.408    |
| Outdegree distribution | 0.000    | 0.172    |

## 10. Discussion and Limits

We modeled best-friend nominations in classroom 09f across three waves (V, X, Y) using a stochastic actor-oriented model (SAOM). We tested (H1) sex homophily (sameX), (H2) transitivity (transTrip), and (H3) popularity reinforcement (inPop). The final model converged successfully, and we interpret the estimates as evidence about mechanisms of tie formation and maintenance in the best-friends network. H1 (sex homophily): The same-sex effect (sameX) is positive and borderline significant (Estimate = 0.246, SE = 0.124;  $t = 1.981$ ), indicating that pupils are somewhat more likely to nominate same-sex peers as best friends over time. This

provides moderate support for sex homophily in best-friend nominations. H2 (transitivity): Transitive triplets are strongly positive (Estimate = 0.466, SE = 0.052;  $t = 8.988$ ), supporting the hypothesis that best-friend ties form through triadic closure (“friends of friends become friends”). This suggests that friendship dynamics are strongly clustered and not independent across dyads. H3 (popularity reinforcement): Contrary to the hypothesis, the indegree-related popularity effect (inPop) is negative (Estimate =  $-0.336$ , SE = 0.079;  $t = -4.274$ ). Conditional on the other effects in the model, pupils who already receive many nominations are less likely to receive additional nominations over time. A plausible interpretation is that “best-friend” nominations are selective and constrained, so nominations may spread across classmates rather than accumulating on a small set of highly nominated pupils.

Other structural effects and controls: Reciprocity is large and positive (Estimate = 3.604, SE = 0.586;  $t = 6.150$ ), indicating that best-friend nominations often become mutual. The outdegree (density) parameter is negative (Estimate =  $-0.814$ , SE = 0.430;  $t = -1.895$ ), reflecting the sparse nature of best-friend networks. Outdegree activity (outAct) is positive (Estimate = 0.127, SE = 0.043;  $t = 2.929$ ), suggesting heterogeneity in sending nominations: some pupils are consistently more active in nominating peers. In contrast, reciprocity-related activity (reciAct) is negative (Estimate =  $-0.460$ , SE = 0.109;  $t = -4.218$ ), indicating that pupils with more reciprocated ties show reduced additional outgoing activity, consistent with constraints/settling into mutual best-friend relations.

Model fit. Goodness-of-fit results improved after updating the model (Model 2): Monte Carlo p-values increased from 0.096 to 0.408 for the indegree distribution and from 0.000 to 0.172 for the outdegree distribution, indicating a substantially better fit (especially for outdegree) while still leaving room for further refinements.

Limitations and improvements: The SAOM estimation requires a constant node set across waves, so the analysis is based on pupils present in all three waves, which reduces sample size and may limit generalizability. Tie stability between waves is not very high, indicating substantial turnover in nominations; richer specifications and additional covariates may be needed to capture this dynamic. Also, popularity is modeled endogenously via indegree dynamics (inPop); future work could incorporate additional exogenous student attributes (from the codebook) to clarify why popularity does not accumulate in best-friend nominations in this classroom. Finally, only a small set of mechanisms was modeled; adding more covariates or comparing multiple classrooms could improve robustness. The results strongly support transitivity and reciprocity in best-friend dynamics, provide moderate evidence for sex homophily, and do not support popularity reinforcement (inPop is significantly negative) for this classroom and network definition.